

Tutorial - 9

Cross Validation Technique:

$$\frac{20}{5} = 4 \quad (20) \rightarrow 5 \text{ fold } K\text{-fold}$$

(16, 4) CN.
Fold - 1

12 $\rightarrow$ data 4 $\rightarrow$ folds.

$$\frac{12}{4} = 3 \quad (9, 3) \rightarrow \text{validation.}$$

Training

KNN (K=5)

① Distance for (46, 84, 3)

(46, 84, 3)

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

$$(46, 84, 3) (25, 50, 8) = \sqrt{(46-25)^2 + (84-50)^2 + (3-8)^2}$$

$$\textcircled{1} 40.28$$

$$4.12 \vee N.$$

$$36.60$$

$$4.24 \vee N.$$

$$29.83 \vee Y.$$

$$2.83 \vee N. \quad 5\text{-Nearest}$$

$$37.11$$

$$4.24 \vee N.$$

$$32.45$$

$$\therefore N = 4.$$

$$Y = 1$$

Prediction = N.

② Distance for (27, 54, 8).

4.41 ✓ Y.

32.02

1.73 ✓ Y.

39.08

7.00 ✓ Y.

25.36

2.24 ✓ Y

39.87

4.58 ✓ Y.

Probability

(8, 07, 25) of (8, 07, 25)

Y = 5

N = 0

Prediction = Y.

③ Distance for (49, 56, 2).

93.68

7.28 ✓ N

38.10

1.41 ✓ N.

33.18 ✓ Y.

4.24 ✓ N.

41.49

2.23 ✓ N.

35.11

(022, 85) N=4

Y = 1

Prediction = N.

Validation Point

Actual

Predicted

(46, 84, 3)

(27, 54, 8)

(49, 86, 2)

Y

N

Y

N

∴ Accuracy = 100%

① Distance Finding

Distance for (46, 84, 3)

① (46, 84, 3) to (25, 50, 8)

$$= \sqrt{(46-25)^2 + (84-50)^2 + (3-8)^2}$$

$$= 40.29$$

② (46, 84, 3) to (45, 80, 3)

$$= \sqrt{(46-45)^2 + (84-80)^2 + (3-3)^2}$$

$$= 4.123$$

③ (46, 84, 3) to (28, 55, 9)

$$= \sqrt{(46-28)^2 + (84-55)^2 + (3-9)^2}$$

$$= 34.65$$

④ (46, 84, 3) to (50, 85, 2)

$$= \sqrt{(46-50)^2 + (84-85)^2 + (3-2)^2}$$

$$= 4.24$$

⑤ (46, 84, 3) to (30, 60, 10)

$$= \sqrt{(46-30)^2 + (84-60)^2 + (3-10)^2}$$

$$= 4.24$$

⑥ (46, 84, 3) to (48, 82, 3)

$$= \sqrt{(46-48)^2 + (84-82)^2 + (3-3)^2}$$

$$= 2.82$$

$$⑦ (46, 84, 3) \text{ to } (26, 52, 8)$$

$$= \sqrt{(46-26)^2 + (84-52)^2 + (3-8)^2}$$

$$= 30.06$$

$$⑧ (46, 84, 3) \text{ to } (47, 88, 2)$$

$$= \sqrt{(46-47)^2 + (84-88)^2 + (3-2)^2}$$

$$= 4.2$$

$$⑨ (46, 84, 3) \text{ to } (29, 58, 9)$$

$$= \sqrt{(46-29)^2 + (84-58)^2 + (3-9)^2} = 31.6$$

Distance for (27, 54, 8)

$$① (27, 54, 8) \text{ to } (25, 50, 8)$$

$$= \sqrt{(27-25)^2 + (54-50)^2 + (8-8)^2} = 4.47$$

$$② (27, 54, 8) \text{ to } (45, 80, 3)$$

$$= \sqrt{(27-45)^2 + (54-80)^2 + (8-3)^2} = 32.01$$

$$③ (27, 54, 8) \text{ to } (28, 55, 9)$$

$$= \sqrt{(27-28)^2 + (54-55)^2 + (8-9)^2} = 1.73$$

$$④ (27, 54, 8) \text{ to } (50, 85, 2)$$

$$= \sqrt{(27-50)^2 + (54-85)^2 + (8-2)^2} = 39.06$$

$$⑤ (27, 54, 8) \text{ to } (30, 60, 10)$$

$$= \sqrt{(27-30)^2 + (54-60)^2 + (8-10)^2} = 7$$

$$⑥ (27, 54, 8) \text{ to } (48, 82, 3)$$

$$= \sqrt{(27-48)^2 + (54-82)^2 + (8-3)^2} = 35.35$$

$$⑦ (27, 54, 8) \text{ to } (26, 52, 8)$$

$$= \sqrt{(27-26)^2 + (54-52)^2 + (8-8)^2} = 2.236$$

$$⑧ (27, 54, 8) \text{ to } (47, 88, 2)$$

$$= \sqrt{(27-47)^2 + (54-88)^2 + (8-2)^2}$$

$$= 39.89$$

$$⑨ (27, 54, 8) \text{ to } (29, 58, 9)$$

$$\sqrt{(27-29)^2 + (54-58)^2 + (8-9)^2}$$

$$= 4.58$$

⑨ Training samples $K=5$

4.47

32.01

1.713

Distance for (49, 86, 2) to (25, 50, 8)

$$① (49, 86, 2) \text{ to } (25, 50, 8)$$

$$\sqrt{(49-25)^2 + (86-50)^2 + (2-8)^2}$$

$$= 43.68$$

(45, 80, 3)

(28, 55, 9)

(50, 85, 2)

(30, 60, 10)

(48, 82, 3)

(26, 52, 8)

(47, 88, 2)

(29, 58, 9)

$$\textcircled{2} (49, 86, 2) \text{ to } (45, 80, 3)$$

$$\sqrt{(49-45)^2 + (86-80)^2 + (2-3)^2}$$

$$= 7.28$$

$$\textcircled{3} (49, 86, 2) \text{ to } (28, 55, 9)$$

$$\sqrt{(49-28)^2 + (86-55)^2 + (2-9)^2}$$

$$= 38.09$$

$$\textcircled{4} (49, 86, 2) \text{ to } (50, 85, 2)$$

$$\sqrt{(49-50)^2 + (86-85)^2 + (2-2)^2}$$

$$= 1.41$$

$$\textcircled{5} (49, 86, 2) \text{ to } (30, 60, 10)$$

$$= \sqrt{(49-30)^2 + (86-60)^2 + (2-10)^2}$$

$$= 33.17$$

$$\textcircled{6} (49, 86, 2) \text{ to } (48, 82, 3)$$

$$\sqrt{(49-48)^2 + (86-82)^2 + (2-3)^2}$$

$$= 4.24$$

$$\textcircled{7} (49, 86, 2) \text{ to } (26, 52, 8)$$

$$\sqrt{(49-26)^2 + (86-52)^2 + (2-8)^2}$$

$$= 41.48$$

$$\textcircled{8} (49, 86, 2) \text{ to } (47, 88, 2)$$

$$\sqrt{(49-47)^2 + (86-88)^2 + (2-2)^2}$$

$$= 2.82$$

$$\textcircled{9} (49, 86, 2) \text{ to } (29, 58, 9)$$

$$\sqrt{(49-29)^2 + (86-58)^2 + (2-9)^2}$$

$$= 35.11$$